

CM300

Power Cutting Mill



Superior Cutting Mill
engineered for rapid
reduction of large particles

- Rapid reduction of large particles from 80 mm to 0,1 mm
- Quick and easy to clean
- Full range of bottom sieves
- Collector volume 5000 ML



The CM300 Cutting Mill can process a large variety of flexible, medium-hard, fiber and mixed products. It can be used for batch and continuous grinding with variable speeds for fast, efficient grinding. At the end of each milling the door can be opened to remove the knives and easily clean the grinding chamber.

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Method of operation

The Model CM300 Cutting Mill is used by Laboratories and processing Companies to granulate solid materials such as waste or wood and thermoplastics. The material to be processed falls into the cutting chamber of the CM300 via a guide chute and is shredded by cutting between rotating and fixed knives until it passes an attached sieve as ground product. The sieve extends over the lower half of the grinding chamber and is easily exchanged. Final particle size is determined by the selected sieve perforation aperture.

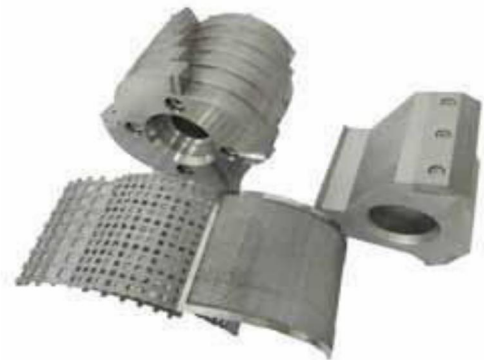
No other Cutting Mill is easier to clean than the CM 300 Cutting Mill. When the grinding process is finished the front door can be opened sideways and the rotor / infeed hopper can be taken out in order to perform a fast and systematic cleaning of the grinding tools.

Features and benefits

- Acoustic Noise Reduction Infeed Hopper made from Bondal. Bondal is a vibration damping composite material with a sandwich structure consisting of a viscoelastic core between two outer steel sheets. Various infeed hopper types are available with wooden/teflon plunger
- Massive grinding chamber made from vibration reducing components
- Solid door with safety switch for easy access and cleaning
- Solid 5 Liters sample collector with
- two side handles and a plexi window
- for sample process observation
- High precision solid steel underframe
- with industrial castor wheels (lockable)



CM300 Cutting Mill open



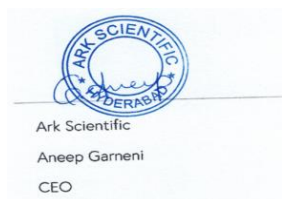
CM300 Cutting bottom sieves and knives



CM300 Cutting bottom hoppers

Technical Specifications – Advanced Composite Cutting Mill (CM300)

No.	Specification Parameter	CM300 – Specification
1	Working Principle	Portable cutting mill should process the sample by strong cutting and shear force. Samples can be crushed many times through shearing action between rotor and cutting blades, and again through shearing action between rotor and sieves. Once sample size is smaller than sieve aperture, product is collected through receiving container.
2	Operation Type	Cutting Type & Cross Beater Type
3	Sample Breaking Time	Less than 1 second
4	Feed Size	60 × 80 mm or smaller
5	Peripheral Speed	Cutting Type: 4–20 m/s or better; Cross Beater Type: 3–25 m/s or better
6	Final Fineness	0.1–4 mm or better. Final fineness controlled by selection of different sieve size.
7	Speed	500–4000 rpm, continuously adjustable
8	Cutting Blade	Manganese steel or equivalent
9	Batch Quantity	100 g or higher
10	Handling Capacity	≥ 30 kg/h
11	Sieves	0.2, 0.5, 1.0, 2.0, 4.0, 6.0, 8.0, 10.0 and 20.0 mm or equivalent size sieves
12	Collector Capacity	0.25–30 L or bigger
13	Grinding Chamber Volume	Stainless steel, 1.1740 steel, hardened steel
14	Grinding Chamber Material	Stainless steel / heavy-metal-free
15	Protection	IP40
16	Noise	<80 dB
17	Rated Power	1.5 kW or more
18	Power Supply	240 V, 50 Hz
19	Instrument Size (W × D × H)	600 × 725 × 1500 mm or less (with stand)
20	Net Weight	120 kg or less
21	Sample Feeding	Samples should be fed from the hopper and then fall into the grinding chamber.
22	Applications	Cutting/shredding of soft, hard, tough and fibrous samples; typical samples include plastic, rubber, plastic-inorganic composites, circuit board, cable, electronic waste, non-metallic waste, etc.



Control, Safety, Accessories & Operating Procedures

Description	CM300 Specification
Control System	Big-screen touch panel / human interface; speed adjustment should be possible by the control panel.
Safety	Electric safety switch to prevent the device from being started when the door is open.
Operation & Cleaning	Instrument should be easy for operation and cleaning. Rotor and sieve should be easily installed, removed and replaced.
Grinding Chamber	Grinding chamber volume should be small and can be cleaned easily.
Cross-Contamination	Grinding set should avoid cross-contamination. Instrument should be supplied with heavy-metal-free material for choice to meet different laboratory requirements.
Low-Heat Grinding	Grinding process should be rapid and produce less heat, suitable for grinding temperature-sensitive materials.
Grinding Insert	Push-button grinding insert should be supplied.
Sample Collection	Sample collection arrangement as specified for the system.
Cyclone Separator	Suitable for handling light or small sample-volume samples; should address heat-transfer problem during grinding and allow easy cleaning.
Universal Feed Hopper	Universal V-type feed hopper.
Cutting Edges	Cutting Type & Cross Beater Type.
Bottom Sieves	Stainless-steel bottom sieves: 0.2, 0.5, 1.0, 2.0, 4.0, 6.0, 8.0, 10.0 and 20.0 mm or equivalent size.
Collecting Bottles	250 ml and 500 ml.
Grinding Inserts	180° and 360° grinding inserts for grinding hard-brittle materials.
Installation & Training	Free installation, commissioning and training for minimum 1 day at buyer's end. Bidder should demonstrate grinding/cutting of different polymer composites and rubber materials during training.
Site Requirements	Bidder should state required space, floor condition and other installation requirements including UPS, dehumidifier, separate special earthing, vibration isolation and compressed air supply, as applicable.
Documentation	Operations and maintenance manual (hard & soft copies); site utilities requirement; certificate that system is brand new and not refurbished/remanufactured; OEM authorization certificate if equipment is sold by authorized dealer.

